

# Wi-Fi RTT Indoor Positioning Using Visibility Matching With NLOS Receptions

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**Abstract**—Round-trip time (RTT)-based Wi-Fi indoor positioning technology is a critical component of the Internet of Things (IoT), enabling various location-based services in smart buildings. However, the positioning accuracy of RTT will be significantly degraded in environments with severe nonline-of-sight (NLOS) reception and a limited number of access points (APs). To solve this challenge, we propose an indoor positioning method based on visibility matching. Compared to the RTT, which is easily affected by NLOS reception, AP visibility is a new positioning feature determined by RTT and signal strength. Additional constraints can be imposed on the positioning system based on the consistency between the observed visibility and the simulated visibility. To improve the accuracy of RTT ranging, a multiwall ranging model is adopted that considers the influence of signal penetration through walls. The proposed positioning method combines the refined RTT ranging result and visibility constraint. It mitigates the impact of NLOS receptions on positioning accuracy through the multiwall ranging model and makes full use of the characteristics of NLOS signal through visibility matching. In different experiments, the proposed method improves median positioning accuracy from 2.3 to 0.64 m compared to other methods.

**Index Terms**—Digital building model, feature matching, indoor positioning, nonline-of-sight (NLOS), Wi-Fi round-trip time (RTT).

## I. INTRODUCTION

THE Internet of Things (IoT) has become a cornerstone of both the technology industry and academic research in recent years. The core of IoT is that users obtain data from the environment through sensors and perform intelligent management of information to enhance operational efficiency [1], [2]. The interaction between smart devices and their environment determines the commands executed by the devices, which directly affects the system's services and user experience. Location-based service (LBS) optimizes resource allocation and space usage by improving energy management efficiency

and enhancing user experience and safety. In the past decade, many IoT applications have implemented LBS in indoor environments such as hospitals, buildings, and shopping centers. For instance, the concept of location of everything (LoE) was proposed in a smart shopping system [3], showcasing the practical applications of positioning systems within IoT [4]. These applications include tracking patients and medical staff in smart hospitals [5], monitoring machines and assets in smart factories [6], indoor navigation in shopping malls [7], and tracking personnel in mines and construction sites [8]. Therefore, the demand for LBS in IoT applications is more urgent than ever.

The global navigation satellite system (GNSS), commonly used outdoors, cannot provide accurate positioning in indoor environments due to signal obstruction. Therefore, many methods have been applied to indoor positioning, such as those based on wireless radio, camera, inertial devices, lidar, etc [1], [9]. Among them, Wi-Fi technology has been widely used because Wi-Fi is a mature telecom infrastructure deployed in almost all buildings, with significant potential for commercialization of IoT applications [10]. In 2016, Wi-Fi IEEE 802.11mc was released, introducing the fine timing measurement (FTM) protocol, also known as the Wi-Fi-round-trip-time (RTT) for ranging. This protocol enables smart devices to estimate their distance to nearby access points (APs) without connection to prevent user privacy. In 2018, Google integrated the Wi-Fi RTT protocol into the newest Android Pie system, attracting many related studies in RTT-based indoor positioning. Common indoor positioning methods, such as fingerprinting [11], [12], ranging [13], and sensor fusion [14], demonstrate higher accuracy and robustness when using RTT.

Despite the advancements in Wi-Fi-based indoor positioning, several challenges hinder its widespread adoption and accuracy. The biggest challenge in indoor positioning is the severe nonline-of-sight (NLOS) reception. Obstacles can block, reflect, or diffract radio signals during propagation, NLOS arise from obstacles (e.g., walls) made of materials with different dielectric constants. The additional ranging errors derive from the altering of the dielectric constant in the propagation (e.g., transmission through walls) or increasing propagation path length (e.g., multipath reflections) [15], [16], [17]. Besides, if the AP quantity is insufficient or the layout is over-concentrated, the accuracy and robustness of the positioning solution will also be impaired [18], [19].

There are several solutions for positioning with NLOS reception: First, use multiple information sources effectively.

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Received signal strength indicator (RSSI), angle of arrival (AOA), or channel state information (CSI) can provide extra positioning characteristics. In [20], a probabilistic method is proposed in the Horus system to estimate position using RSSI data. However, the RSSI values at a fixed location often exhibit significant variations over time. In [21], the researcher explains the undesirable RSSI value and uses CSI to construct the fingerprinting system at the receiver. Arraytrack in [22] uses the multiple signal classification (MUSIC) algorithm to obtain AOA and achieve a decimeter-level, low-latency positioning system. While CSI and AOA require additional hardware design support. Second, mitigate the impact of NLOS on positioning. Liang et al. [17] proposed a wall-penetrating model, which is specifically used to mitigate the NLOS ranging deviation caused by wall-penetrating signal propagation in complex indoor environments. In [23], the accuracy of the free space path loss (FSPL) model under NLOS transmission conditions for high-frequency signals is studied. They mitigate the NLOS ranging error through modeling and use it to calibrate the distance estimation under NLOS receptions. The weighted Taylor series least squares (TS-LS) [19] and the linear programming (LP) approach [18] are proposed by integrating both LOS and NLOS range estimates. They reduce the weight of NLOS ranging to enhance the overall positioning accuracy but do not perform well when the number of APs is insufficient. Finally, leverage the impact of NLOS on positioning. Fingerprinting determines the user's position by analyzing the characteristics of signals in the environment including NLOS signal propagation. Usually, the fingerprinting positioning systems typically rely on RSSI, RTT, and CSI [11], [13], [21], but they tend to be labor-intensive. In [24] and [25], the spatial information from wireless radio signals is leveraged for multipath-assisted positioning. The multipath components (MPCs) reflected off walls are interpreted as LOS signals from a virtual transmitter, which is positioned at the location of the physical transmitter mirrored at the wall.

Coincidentally, GNSS positioning in urban canyons also faces a similar NLOS reception challenge. Buildings can obstruct and reflect satellite signals, resulting in positioning errors exceeding 10 m [26]. Groves [27] and Wang et al. [28] proposed shadow matching, a satellite visibility matching method based on the 3-D building model to refine positioning accuracy. Shadow matching employs the principle of visibility matching to determine a solution by comparing the strength of received signals with the predicted satellite visibility. In early research [27], [29], satellite visibility was defined as a binary value based on a hard threshold derived from the carrier-to-noise density power ratio ( $C/N_0$ ). For each candidate point, the matching degree between predicted and measured visibility is represented by a matching score. The final solution is obtained by taking a weighted average of the coordinates of several high-scoring candidate points. Subsequent studies [26], [30], [31] have shown that Bayesian theory can be used to optimize the assessment of satellite visibility, leading to improved positioning results. In the across-street positioning experiment, the positioning accuracy based on shadow matching was improved from more than 10 m to less than 3 m [26]. Wi-Fi and satellite signals are

electromagnetic waves operating at different frequencies and modulation modes. In our case, like satellite signals, Wi-Fi signal attenuation can also be used to determine visibility when the signals are blocked. Therefore, shadow matching has the potential to be applied to indoor positioning. It can avoid the large positioning error introduced by outliers in NLOS environments and enhance the robustness and accuracy of the positioning system.

To advance the automation of intelligent devices in indoor environments, optimize products and services by analyzing user behavior, and efficiently help users quickly find the resources or services they need in the IoT landscape. We should overcome the challenges of NLOS indoor positioning and provide accurate positioning services. We introduce a Wi-Fi RTT indoor positioning method using visibility matching. First, we refine RTT ranging using a multiwall polynomial model for positioning. Next, we divide the positioning area into a grid and compute the simulated visibility of each AP within each grid cell. We then calculate the observed visibility of each AP to the user based on the fitted LOS propagation model. Finally, we identify the most probable user locations within the area using refined RTT ranging and visibility matching. To the best knowledge, this is the first paper to introduce AP visibility and the RTT multiwall ranging model in traversal solving for indoor positioning, achieving the utilization of advantageous features of NLOS and the mitigation of error terms introduced by NLOS. The contributions of this article are as follows.

- 1) Propose a location constraint method based on AP's visibility matching. This method fully leverages the characteristics of NLOS signals and introduces a new feature from APs to impose additional constraints on positioning.
- 2) Propose an integrated positioning method utilizing refined RTT ranging with a multiwall model and visibility matching constraints. This method not only mitigates the adverse effects of NLOS on positioning but also leverages the characteristics of NLOS signals in indoor NLOS environments.
- 3) Validate the effectiveness of the proposed algorithm using a diverse set of self-collected data from various real-world environments. A thorough assessment of the algorithm's performance is conducted using multiple evaluation metrics, including ranging accuracy, positioning accuracy, and cumulative distribution function (CDF).

In Section II, we introduce the traditional ranging and positioning model. Section III models the RTT ranging by considering the effect of multiple walls. Section IV discusses the visibility matching. Section V evaluates the feasibility of our method in indoor scenarios. Section VI presents the conclusion and outlines potential directions for future work.

## II. MODELS FOR RANGING AND POSITIONING

The radio-based indoor positioning results are closely related to the ranging results. This section briefly introduces the ranging method based on RTT or RSSI, as well as the position estimation method based on the ranging results.

### A. Models for RSSI and RTT Ranging

RSSI ranging measurement models use the strength of received signals to estimate the distance between devices. In wireless communication, signals attenuate as they propagate due to distance, obstacles, and environmental factors. There is a relationship between the measured RSSI and the propagation distance. The attenuation of signal strength can be described by FSPL models [32], [33]

$$s = 10N \cdot \log_{10} d + A \quad (1)$$

where  $s$  indicates the estimated RSSI based on the distance between the receiver and the transmitter.  $N$  indicates the path loss coefficient and  $A$  represents the RSSI value when  $d = 1$ , both of which are precalibrated values.

Theoretically, the RTT measurement value should represent the true distance between the receiver and the transmitter. However, factors such as network delay, clock synchronization error, and multipath effect can cause differences between the RTT measurement and the true distance. Even in an antenna measurement room without multipath effect, the ranging error is about 1 m [13]. In indoor positioning applications, ranging error is closely related to the hardware and environment. Therefore, we use second-order polynomial fitting to correct the RTT positioning results to improve positioning accuracy

$$r = c_0 + c_1 \cdot d + c_2 \cdot d^2 \quad (2)$$

where  $r$  represents the RTT ranging;  $d$  indicates the distance from the receiver to the transmitter.  $c_0, c_1, c_2$  are the second-order polynomial coefficients. However, indoor positioning has a large number of NLOS receptions, and whether the signal is visible will affect the RTT ranging and the accuracy of the fitting model. Therefore, in the next section, we employ the RTT polynomial fitting model considering multiple walls.

### B. Models and Parameters for Positioning

This study aims to know the location of the user  $\mathbf{p}_k \in \mathbb{R}^2$ . We define RTT ranging as  $\mathbf{r}_k \triangleq \{r_k^j | r_k^j \in \mathbb{R}, j = 1, \dots, J\}$ , define RSSI measurement as  $\mathbf{s}_k \triangleq \{s_k^j | s_k^j \in \mathbb{R}, j = 1, \dots, J\}$ , and define APs' location as  $\mathbf{A} \triangleq \{\mathbf{a}^j | \mathbf{a}^j \in \mathbb{R}^2, j = 1, \dots, J\}$ .  $k$  is the time series, and  $j$  is the AP sequence. The user's position can be determined by

$$\mathbf{p}_k = f(\mathbf{A}, \mathbf{r}_k). \quad (3)$$

$\mathbf{p}_k, \mathbf{A}$ , and  $\mathbf{r}_k$  satisfy

$$\begin{aligned} \|\mathbf{p}_k - \mathbf{a}^1\| &= r_k^1 \\ &\vdots \\ \|\mathbf{p}_k - \mathbf{a}^J\| &= r_k^J \end{aligned} \quad (4)$$

where  $\|\cdot\|$  indicates the Euclidean distance between point  $\mathbf{p}_k$  and point  $\mathbf{a}^j$ . This method is trilateration, which can be converted into a least squares problem and is the fundamental principle of ranging positioning. The weighted least squares is applied in trilateration because inaccurate ranging derived from NLOS reception will impair the positioning accuracy [19]. However, when the number of APs is insufficient,

the robustness of weighted least squares may weaken and even degrade to ordinary least squares.

We introduce the RTT multiwall ranging model to reduce the ranging bias, and introduce the floorplan to leverage NLOS information. In the following content, we will solve this equation:

$$\mathbf{p}_k = f(\mathbf{p}_{k-1}, \mathbf{A}, \mathcal{F}, \mathbf{r}_k, \mathbf{s}_k) \quad (5)$$

where floor plan  $\mathcal{F} \triangleq \{\mathbf{F}_l = (\mathbf{f}_l^1, \mathbf{f}_l^2) | \mathbf{f}_l^1, \mathbf{f}_l^2 \in \mathbb{R}^2, l = 1, \dots, L\}$  are known. The building's floor plan is digitized as an array of wall segment tuples, with each tuple  $\mathbf{F}_l$  representing the start and end position of a wall.

## III. RTT MULTIWALL RANGING MODEL

### A. Model Description

The RTT multiwall ranging model can be expressed as

$$\hat{r}_k^j = h(\mathbf{p}_k, \mathbf{a}^j, \mathcal{F}, \mathbf{c}^j) \quad (6)$$

where  $\hat{r}_k^j$  is the estimated RTT ranging;  $\mathbf{c}^j$  is the coefficient set of the polynomial. Under NLOS propagation, the direct path of the electromagnetic wave is severely attenuated, and the reflection paths introduce ranging errors. In addition, the propagation and refraction of dielectric electromagnetic waves in different media will also impair the ranging results [34]. These side effects are related to the dielectric constant and thickness of the wall [17]. To quantify the impact of walls on electromagnetic wave transmission, we introduce an error term  $\Delta_b$  to define the multiwall RTT ranging error model  $h(\mathbf{p}_k, \mathbf{a}^j, \mathcal{F}, \mathbf{c}^j)$

$$\hat{r}_k^j = h(\mathbf{p}_k, \mathbf{a}^j, \mathcal{F}, \mathbf{c}^j) = c_0^j + c_1^j \cdot d_k^j + c_2^j \cdot d_k^{j^2} + \sum_{b \in \mathcal{V}^j} \Delta_b \quad (7)$$

where  $d_k^j = \|\mathbf{p}_k - \mathbf{a}^j\|_2$  indicates the Euclidean distance from user to  $j$ th AP;  $\mathbf{C} \triangleq \{\mathbf{c}^j | \mathbf{c}^j = \{c_0^j, c_1^j, c_2^j\}, j = 1, \dots, J\}$  is the set of second-order polynomial coefficients associated with  $j$ th AP. Here, we define  $\mathcal{V}^j$  to represents the indices of walls crossed by the direct path between  $j$ th AP and the user,  $\mathcal{V}^j = \emptyset$  if the signal is LOS propagation.  $\Delta_b$  is the time delay introduced by passing through the wall, different by wall properties.

### B. Parameters Fitting

The application of the RTT multiwall ranging model starts from the fitting of parameters, as shown in Algorithm 1. We need to fit  $|\mathbf{c}| = 3J + 1$  parameters, including second-order polynomial coefficients for  $J$  APs and the passing through error  $\Delta_b$ . The function *isintersect* determines whether the segment line between the user and the AP intersects with the wall by using geometric intersection methods. The cost function  $\mathcal{J}(\mathbf{c}^j)$  represents the two-norm of all the losses at epoch  $k$  for AP  $j$ , defined as

$$\begin{aligned} \mathcal{J}(\mathbf{c}^j) &= \sum_{k=1}^K \sum_{j=1}^J \mathcal{L}(r_k^j, \hat{r}_k^j) \\ &= \sum_{k=1}^K \sum_{j=1}^J \mathcal{L}(r_k^j, h(\mathbf{p}_k, \mathbf{a}^j, \mathcal{F}, \mathbf{c}^j)). \end{aligned} \quad (8)$$

**Algorithm 1** Parameters Fitting of Multiwall Ranging Model for AP  $j$ 
**Input:**

- $\{\mathbf{p}_k\}$ : the location of the user.
- $\{r_k^j\}$ : the RTT ranging.
- $\{\mathbf{a}^j\}$ : the location of APs.
- $\{\mathbf{F}_l\}$ : wall position set.
- $\mathbf{c}_0^j$ : initial polynomial coefficient in multi-wall model.

**Output:**

- $\mathbf{c}^{j*}$ : polynomial coefficient in multi-wall model.
- 1: Build and initialize the loss function  $\mathcal{J}(\mathbf{c}_0^j) = 0$ .
- 2: **for all**  $k = 1, 2, \dots, K$  **do**
- 3:   **for all**  $l = 1, 2, \dots, L$  **do**
- 4:     **if**  $\text{isintersect}(\mathbf{F}_l, \mathbf{p}_k - \mathbf{a}^j) = 1$  **then**
- 5:        $\mathcal{V}_k^j \leftarrow \mathcal{V}_k^j \cup l$
- 6:     **end if**
- 7:      $d_k^j \leftarrow \|\mathbf{p}_k - \mathbf{a}^j\|$
- 8:      $\hat{r}_k^j \leftarrow c_0^j + c_1^j \cdot d_k^j + c_2^j \cdot d_k^j{}^2 + \sum_{b \in \mathcal{V}^j} \Delta_b$
- 9:      $\mathcal{J}(\mathbf{c}^j) \leftarrow \mathcal{J}(\mathbf{c}^j) + \mathcal{L}(r_k^j, \hat{r}_k^j)$
- 10:   **end for**
- 11: **end for**
- 12:  $\mathbf{c}^{j*} = \arg \min_{\mathbf{c}^j \in \mathbb{R}^{3J+1}} (\mathbf{c}^j)$
- 13: **return**  $\mathbf{c}^{j*}$ .

Here,  $\mathcal{L}(\cdot)$  is the Huber loss function, which penalizes the weight of outliers and noisy data. For initial setting  $\mathbf{c}_0$ , we let  $c_2^0 = 0$ ,  $c_1^0 = 1$ ,  $c_0^0 = 0$  and  $\Delta_b^0 = 1$ .

### C. Application of RTT Multiwall Model

A basic and stable trilateration positioning system requires a valid trilateration structure formed by three noncollinear APs that are adequately spaced. The positioning accuracy is positively correlated with the number of APs and the ranging accuracy of each AP [35], [36], [37]. Our research uses the RTT ranging model to mitigate the ranging error. Then, we use trilateration to initially obtain the possible location of the user.

We divide the venue into many grids, each grid potentially represents the user's location. The grids are referred to as candidates, defined as  $\mathbf{E} = \{\mathbf{e}_i \in \mathbb{R}^2 | i = 1, \dots, I\}$ ,  $i$  indicates the candidates' index. In our model, the user's location is in a circle with AP  $j$  as the center and radius  $\hat{r}_k^j$ , which is referred to as AP constraint. The binary AP constraints  $P_i^j$  imposed on the candidates are defined as follows:

$$P_i^j = \begin{cases} 1, & \left| \|\mathbf{e}_i - \mathbf{a}^j\| - |\hat{r}_{i,k}^j + \delta^j| \right| \leq g(\text{w constrain}) \\ 0, & \left| \|\mathbf{e}_i - \mathbf{a}^j\| - |\hat{r}_{i,k}^j + \delta^j| \right| > g(\text{w/o constrain}) \end{cases} \quad (9)$$

where  $g$  denotes the tolerance in meters;  $\delta^j \in \mathcal{N}(\mu_{\text{con}}^j, \sigma_{\text{con}}^j{}^2)$  represents the measurement noise, following a Gaussian distribution. The AP constraint vector is defined as

$$\mathbf{P}_i = [P_i^1, \dots, P_i^J]^\top. \quad (10)$$

Fig. 1 illustrates the AP constraints in a 3-AP positioning system. The figure shows three rings centered at the AP

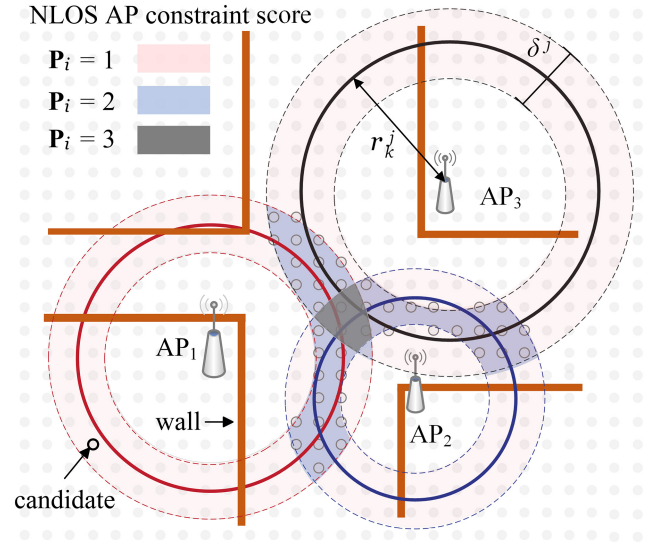


Fig. 1. AP constraints of the candidates in a 3 AP positioning system. The area is divided into many candidates. Each candidate has the corresponding AP constraint score  $\mathbf{P}_i$  based on RTT measurement value  $r_k^j$ .

locations to represent the ranging results of the RTT, including the predicted distance  $\hat{r}_{i,k}^j$  and error scope  $\delta^j$ . Equation (9) determines whether candidate  $i$  is within the constraint ring of AP  $j$ . To quantify the AP constraints on each candidate, we use the one-norm as the AP constraints score for each candidate  $\mathbf{e}_i$ , which is defined as

$$P_i = \|\mathbf{P}_i\|_1. \quad (11)$$

The more constraints a candidate satisfies, the higher its possibility of being the users' location. For the robustness and accuracy of the solution, the tolerance  $g$  needs to be chosen carefully based on experience. If  $g$  is overlarge, the constraints of the positioning solution will not be sufficient and the positioning accuracy will be reduced. If  $g$  is too small, it may fall into a local optimal solution, resulting in no intersection between multiple AP constraints. We define AP constraint scores greater than 1 as valid constraints, and the corresponding candidate set as  $\Omega_{va} = \{\mathbf{e}_i | P_i > 1\}$ . The selection of tolerance  $g$  should tradeoff the following conditions: 1) the highest AP constraint score corresponding to the elements in  $\Omega_{va}$  is equal to the number of APs; 2) the region composed of the elements in  $\Omega_{va}$  is continuous; and 3) the area formed by the elements of  $\Omega_{va}$  is minimized.

### IV. POSITIONING METHOD BASED ON VISIBILITY MATCHING

To mitigate the positioning error indoors, a visibility matching method based on virtual simulated visibility and the user's observed visibility is proposed in this section. We demonstrate how to estimate user location using visibility matching and AP constraints. The flow chart of the proposed algorithm is shown in Fig. 2.



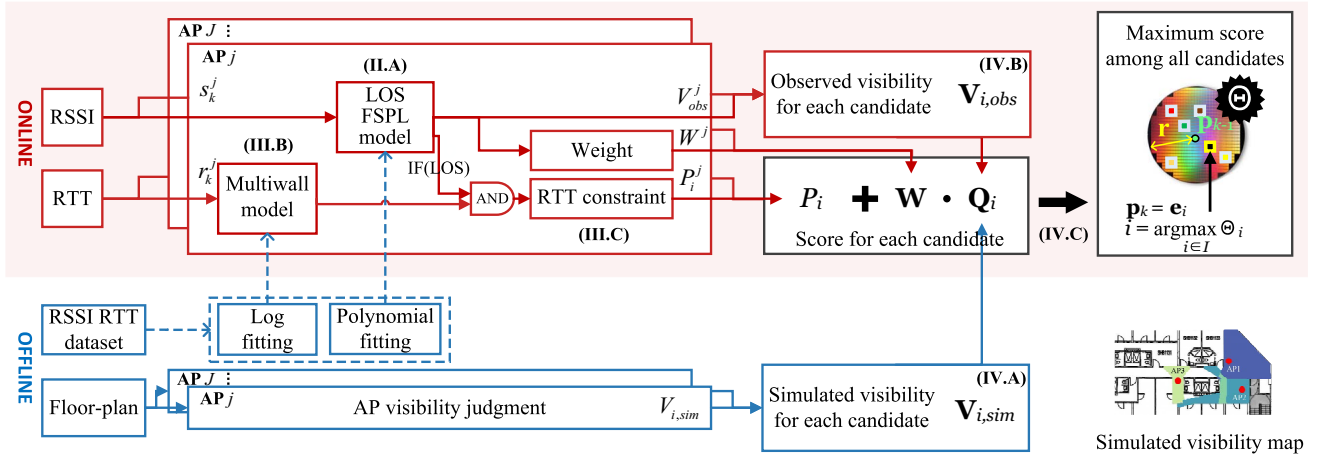


Fig. 2. Flowmap of the proposed algorithm in this article. The blue module represents offline calculations. The red module represents online calculations. The multiwall ranging model and LOS logarithmic model are used online but require offline fitting.

### A. Simulated Visibility

The simulated visibility is constant, it depends on whether the position of AP  $\mathbf{a}^j$  is visible from the sight of the candidates  $\mathbf{e}_i$ . If the line segment between the candidate and AP intersects the wall, the AP is invisible and the signal NLOS reception at  $\mathbf{e}_i$ . We define the simulated visibility of candidate  $\mathbf{e}_i$  and AP  $\mathbf{a}^j$  as

$$V_{i,\text{sim}}^j = \begin{cases} 0, & \forall \mathbf{F}_l \in \mathcal{F}, \text{ is\_intersect}(\mathbf{F}_l, \mathbf{e}_i - \mathbf{a}^j) = 1 \\ 1, & \forall \mathbf{F}_l \in \mathcal{F}, \text{ is\_intersect}(\mathbf{F}_l, \mathbf{e}_i - \mathbf{a}^j) = 0. \end{cases} \quad (12)$$

The function  $\text{is\_intersect}(\cdot)$  determines whether the line segment connecting the candidate and the AP intersects with any wall segment by using geometric intersection judgment methods. If  $\text{is\_intersect}(\cdot) = 1$ , the AP's signal is classified as NLOS reception; if  $\text{is\_intersect}(\cdot) = 0$ , the AP's signal is classified as LOS reception. For candidate  $i$ , the visibility score is

$$\mathbf{V}_{i,\text{sim}} = [V_{i,\text{sim}}^1, \dots, V_{i,\text{sim}}^J]^\top. \quad (13)$$

Fig. 3 illustrates the spatial distribution of simulated visibility affected by four APs, the simulated visibility of all candidates in the gray area is  $\mathbf{V}_{i,\text{sim}} = [1, 1, 1, 0]$  when only AP4 is invisible.

### B. Observed Visibility

RSSI indicates the signal strength, related to the ranging distance and signal attenuation. Theoretically, the propagation distance can be estimated based on RSSI in (1), but the result is generally inaccurate because the wall will block and attenuate the signal. On the other hand, we can also determine whether the AP is visible based on the strength attenuation property when the signal penetrates walls. If RSSI and RTT satisfy the fitted logarithmic correlation, the signal is LOS propagation.

In the illustration of RSSI versus RTT in Fig. 4, we define  $E_u$  and  $E_l$  as the upper and lower bound of RSSI, representing the maximum and minimum values of the fitted LOS logarithmic model, respectively. All signals have been divided into three categories: LOS, NLOS, and uncertain. In

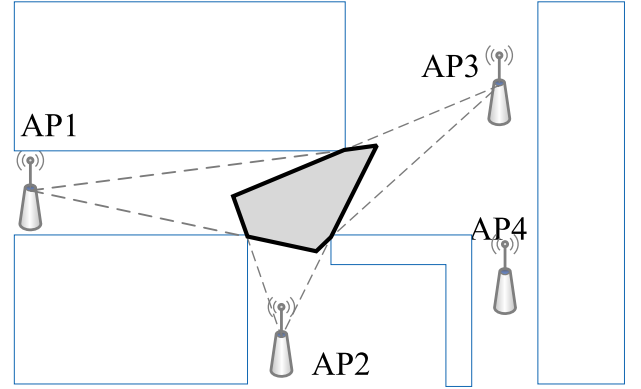


Fig. 3. Illustration of visibility matching constraint area. The blue-filled areas are walls. All candidate points in the gray area have the same simulated visibility.

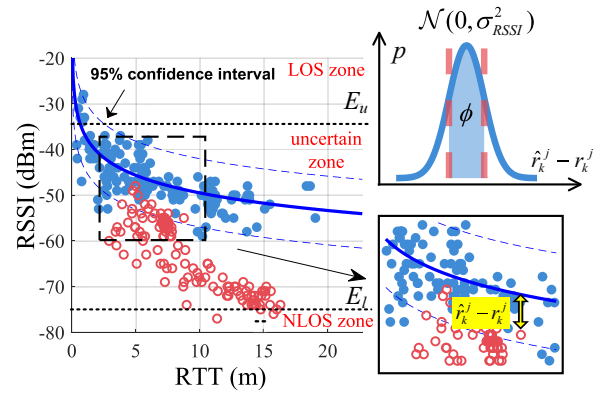


Fig. 4. Illustration of RSSI versus RTT. LOS, NLOS, and uncertain zone are divided by  $E_u$  and  $E_l$ . The blue points represent LOS tuples, and the red points represent NLOS tuples. The solid blue line is the fitted LOS FSPL model, while the dashed blue line indicates the 95% confidence interval. The subplot in the lower right shows the residuals of the predicted RSSI values; the subplot in the upper right displays the definition of  $\phi$ .

the LOS zone,  $s_k^j > E_u$ ; in the NLOS zone,  $s_k^j < E_l$ ; in the uncertain zone,  $E_l \leq s_k^j \leq E_u$ . To quantify the consistency of the signal to the LOS logarithmic model, we introduce the

confidence coefficient

$$V_{\text{obs}}^j = \begin{cases} 1, & (r_k^j, s_k^j) \in \Omega \\ 0, & (r_k^j, s_k^j) \notin \Omega, \end{cases}$$

$$W^j = \begin{cases} 1, & s_k^j > E_u \\ \phi, & (E_l < s_k^j < E_u) \wedge (V_{i,\text{sim}}^j = 1) \\ 1 - \phi, & (E_l < s_k^j < E_u) \wedge (V_{i,\text{sim}}^j = 0) \\ 1, & s_k^j < E_l. \end{cases} \quad (14)$$

For candidate  $i$ , the observed visibility score and confidence coefficient are

$$\mathbf{V}_{i,\text{obs}} = [V_{i,\text{obs}}^1, \dots, V_{i,\text{obs}}^J]^\top$$

$$\mathbf{W} = [W^1, \dots, W^J]^\top. \quad (15)$$

Here,  $\Omega$  represents the 95% confidence interval of the LOS fitting model, the signal tuple  $(r_k^j, s_k^j)$  within  $\Omega$  is considered to be consistent with the LOS fitting model,  $V_{\text{obs}}^j = 1$ , otherwise  $V_{\text{obs}}^j = 0$ . We assume that the residual  $|r_k^j - r_k^j|$  follows a Gaussian distribution  $\mathcal{N}(0, \sigma^2_{\text{RSSI}})$ ,  $\phi$  denotes the integral of the residual probability density function. In the LOS zone or NLOS zone (i.e.,  $r_k^j < E_l$  or  $r_k^j > E_l$ ), we consider the distinction between LOS and NLOS to be accurate, with  $W^j = 1$ ; in the uncertain zone (i.e.,  $E_l < r_k^j < E_u$ ), for the consistency of the logical operations of observed and simulated visibility, we define two attributes NLOS consistency and LOS consistency whose sum is 1 to describe distinction confidence, with  $0 < W^j < 1$ .

### C. Positioning Method Based on Visibility Matching

In this part, we establish a score criterion for the user's potential location.  $\mathcal{O}_I$  is a circular area with a radius of  $R$  and the user's last-epoch location as the center,  $\mathcal{I}$  is the index of candidates belong to  $\mathcal{O}_I$ . For any candidate  $\mathbf{e}_i \in \mathcal{O}_I$ , we define the visibility matching score  $Q_i^j$  between the simulated and observed visibility

$$Q_i^j = \text{XNOR}(V_{\text{obs}}^j, V_{i,\text{sim}}^j). \quad (16)$$

*XNOR* is exclusive NOR, it returns 1 when inputs  $V_{\text{obs}}^j$  and  $V_{i,\text{sim}}^j$  are identical. The visibility matching vector  $\mathbf{Q}_i \in \mathbb{R}^J$  for candidate  $i$  is defined as

$$\mathbf{Q}_i = [Q_i^1, \dots, Q_i^J]^\top. \quad (17)$$

Given the confidence coefficient  $\mathbf{W}$ , we define the influence of visibility matching on candidate  $i$ , denoted as  $Q_i$

$$Q_i = \mathbf{W}^\top \mathbf{Q}_i. \quad (18)$$

We have already defined the AP constraint score  $P_i$  derived from the overlap of the constraint rings in (9). Here, we sum AP constraints and visibility constraints as the final candidate score  $\Theta_i$ , which is defined as

$$\Theta_i = Q_i + P_i. \quad (19)$$

Then, the candidate  $i$  with the maximum  $\Theta_i$  is considered as the user's location, we have

$$\mathbf{p}_k = \mathbf{e}_i, \quad \text{where } i = \arg \max_{i \in \mathcal{I}} \Theta_i. \quad (20)$$

When multiple solutions exist, that is, several  $\mathbf{e}_i$  correspond to the same maximum  $\Theta_i$ , their average position is calculated. This method utilizes the user's mobility, AP positioning, and visibility constraints. Its performance tends to be fully exploited in locations where visibility changes frequently.

## V. EVALUATION

### A. Experiment Setup

To evaluate the performance of the proposed algorithm, we conducted experiments in the student dormitory at PolyU, HK. The experiments were conducted in three different environments: 1) a narrow corridor; 2) a wide corridor; and 3) a suite.

- 1) *Narrow Corridor (Straight-Line Motion Experiment)*: The corridor had a width of approximately 1.5 m and a length of 10 m. This environment was chosen to verify the algorithm's performance in a narrow indoor environment. Three APs were set up in the experimental scene. Most of the experimental trajectories were located in the area of signal LOS propagation, and a few were located in the area of NLOS propagation. The walls of the corridor were made of concrete.
- 2) *Spacious Corridor (Closed Loop Experiment)*: This place was the square area end of the narrow corridor, with dimensions of 5 m in width and 5 m in length. This setting was chosen to verify the performance of the algorithm when the user moves back and forth over. Two APs were set up within the suite, one AP was set up in the spacious corridor, providing the environment with at least one visible AP.
- 3) *Suite (AP Hybrid Visibility Experiment)*: This environment was chosen to evaluate the algorithm's performance in a complex indoor setting with hybrid AP visibility. The complex layout, severe multipath caused by furniture and clutter, and the movement of experimenters in different rooms more accurately reflect real-world usage.

The experiment site and equipment are shown in Fig. 5(a) and (b). Each experiment contains three Google Nest Wi-Fi APs that support RTT measurement with 80 MHz bandwidth at 5 GHz. The data packets were sampled at 3 Hz, including timestamps, RTT, standard deviation of RTT, and RSSI. The ground truth was obtained using a LiDAR Livox Mid-360 through Fast-lio [38]. The ground truth sampling frequency is 10 Hz, including timestamps, and relative change in position. The RTT timestamps are used as a reference on data alignment between LiDAR and RTT.  $r$  is set to 10 m of sampling area when looking for the initial position, and takes the average of the positioning results at 10 epochs. After the initial value is determined,  $r = 1$  m, based on user movement constraint (velocity  $\leq 1$  m/s), to minimize the interference in the score map.

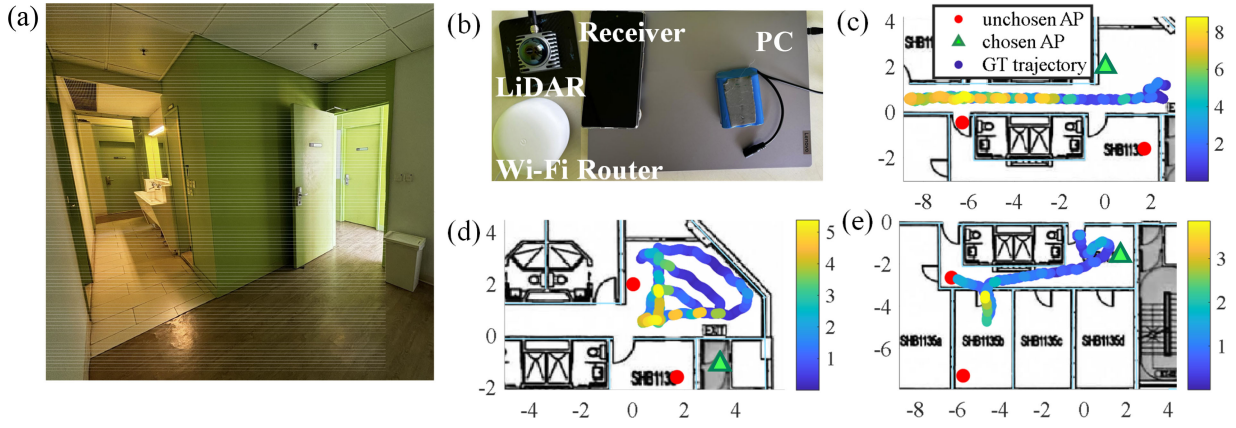


Fig. 5. Experiment setting and map of indoor ranging errors along the GT trajectory in three experiments. The background shows the floor plan of the indoor environment. Colored dots represent ranging errors at different positions along the GT trajectory, with the color scale indicating the magnitude of the error in meters. Red circles denote unchosen APs, and the green triangle represents the chosen AP utilized for visualizing ranging errors. (a) Experiment site. (b) Experimental equipment. (c) Ranging error in narrow corridor experiment (taking AP 1 as an example). (d) Ranging error in spacious corridor experiment (taking AP 3 as an example). (e) Ranging error in suite experiment (taking AP 1 as an example).

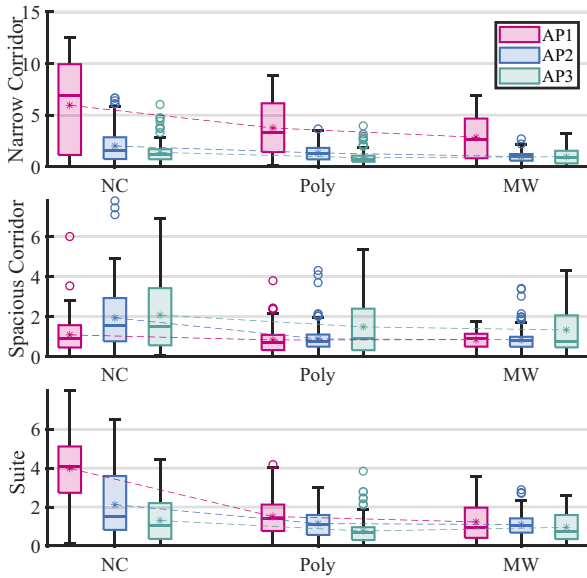


Fig. 6. Boxplots depict the absolute range prediction errors (in meters) for NC, Poly, and MW across 3 APs in three experiments. The upper and lower hinges represent the 75th and 25th percentile errors, respectively. Whiskers extend to the 95th and 5th percentile errors, with the median error shown as the middle bar.

### B. Multiwall Ranging Model Evaluation

To obtain the parameters for the multiwall ranging model, we fixed the AP and measured the range to its location while walking around the room with LiDAR and RTT receiver. We ensured that the distance from the receiver to the AP remained between 1 and 10 m. The following data would be recorded: 1) the location of the AP; 2) the RTT ranging measurement; 3) the true distance from the receiver to the AP calculated based on the relative position changes from the LiDAR; and 4) the number of intersecting walls calculated based on the receiver's true position and the floor plan. These data correspond to  $\mathbf{a}^i$ ,  $r_k^j$ ,  $\hat{r}_k^j$ , and  $|\mathcal{V}|$  in Algorithm 1, respectively.

By applying the multiwall ranging model to the three experiment sites datasets, we obtained the estimated distances

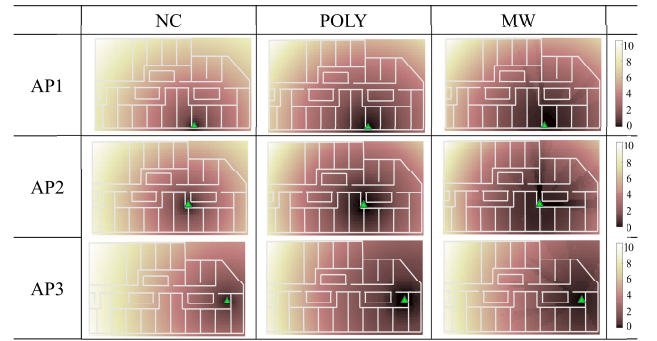


Fig. 7. Error distribution of indoor positioning for different APs using NC, POLY, and MW error model in suite experiment. The color gradient from yellow to dark red indicates the magnitude of the ranging error, with yellow representing larger errors and dark red representing smaller errors. The green mark is the analyzed AP.

from the user to different APs. Boxplot Fig. 6 displays the error of the estimated ranging (in meters) for no correction (NC), polynomial (Poly), and multiwall (MW) models based on data from three APs across three experiments. The mean values are marked with asterisks and connected by dashed lines. MW has a smaller overall error range in all experiments. For instance, in the range performance of AP1 in the narrow corridor experiment, the third quartile and median decreased by approximately 5 m, demonstrating better performance than the polynomial and NC models. In this experiment, AP1's ranging error is relatively greater because AP1 is adjacent to an inaccessible room. When the signal passes through this room, it not only penetrates the walls but also undergoes refraction and reflection within the bathroom, leading to complex multipath effects. As shown in the ranging error map in Fig. 5(a), most larger errors occur when the signal crosses the bathroom. In the spacious corridor experiment in Fig. 5(b), AP3 faces a similar issue because it is placed in a fire escape, where the signal experiences significant attenuation when passing through the heavy fire door.

We also evaluate the spatial distribution of the MW ranging model. We predict the RTT measurements in every corner of

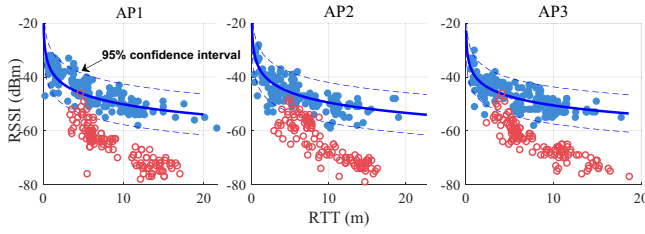


Fig. 8. Distinction of observed visibility among different APs. The red hollow points denote NLOS signals, the blue solid points denote LOS signals, the blue line represents the fitted curve for the LOS signals, and the blue dotted line represents the 95% confidence interval of the curve fitting.

the experiment venue based on the geometry and MW model. Fig. 7 illustrates the error distribution of indoor positioning for three APs using NC, POLY, and MW in the experiment. For the NC model, the ranging error is smaller when close to AP, but it increases rapidly at a distance. For the Poly model, the ranging error is more uniform and smaller than NC. In terms of the MW model, distinct color changes can be observed at the boundaries of the walls. These variations indicate that different numbers of walls through which the signal passes can lead to differences in ranging errors. Overall, the MW ranging model exhibits greater accuracy when compared to the NC and POLY models, especially in areas near the APs and along the boundaries of the walls.

### C. Observed and Simulated Visibility Evaluation

1) *Observed Visibility Distinguishing Model*: The differentiation of the LOS signal and NLOS signal directly determines the quality of the visibility matching. Fig. 8 shows the fitted FSPL model of the LOS signal, with the blue solid points and red hollow points representing the NLOS and LOS signal, respectively. Compared with LOS signals, the attenuation of NLOS signals is more obvious as the distance increases. When the distance is equal to 10 m, the signal strength of NLOS is 15 dBm lower than that of LOS on average. When the distance is less than 10 m, the areas of NLOS and LOS signals overlap a small part. We use the weight coefficient in (14) to mitigate the influence of recognition error. Overall, when RSSI exceeds  $E_u = -40$  dBm, the corresponding signal is classified as LOS; Conversely, when the RSSI remains below  $E_l = -65$  dBm, the corresponding signal is classified as NLOS. For signals belonging to the uncertain zone, i.e.,  $-65$  dBm  $<$  RSSI  $<$   $-40$  dBm, we use weight coefficient  $\mathcal{W}^j$  defined in Section IV-B to adjust the visibility matching result.

2) *Simulated Visibility Map in 20m  $\times$  10m Venue*: In the experiment, we partitioned the experiment area, with 20 m in length and 10 m in width, into a grid of 200  $\times$  100 grids. Each grid  $i$  represents a 0.1 m  $\times$  0.1 m small area, storing its visibility  $V_{i,\text{sim}}^j$  and the number of walls between one grid and APs. In the multiwall model, we consider that the additional distance introduced by NLOS propagation in RTT ranging is from two main factors: 1) the change in propagation time caused by the change in dielectric properties when the signal is blocked by the wall and 2) the additional path length caused by refraction when the signal travels through different media. Therefore, the number of walls is positively correlated with

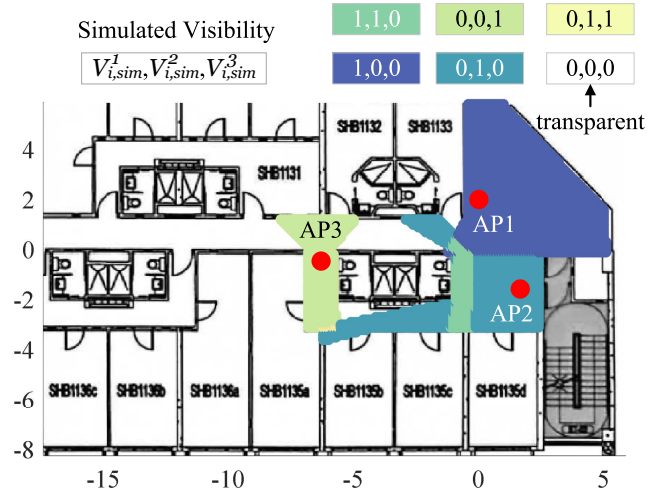


Fig. 9. Illustration of the simulated visibility score map, take the narrow corridor experiment environment as an example. The red dots are the position of APs. Regions with different colors represent different visibility combinations of APs.

the extra distance introduced in RTT measurement. The last term  $\sum \Delta_b$  in (7) illustrates this relationship. Fig. 9 is the simulated visibility score map. The area of the same color has the same simulated visibility score and will receive the same  $\Theta_i$  in visibility matching. This detailed spatial division facilitates a more accurate analysis of the position of users.

### D. Visibility Matching Positioning Method Evaluation

Fig. 10 summarizes the positioning results of different algorithms in the experiment. The following methods are used to compare with the proposed method.

- 1) NC uses raw RTT measurement data without any ranging error mitigation by any RTT ranging model.
- 2) Trilateration (TRI) [18], [19] is the traditional least squares algorithm for the positioning method, it requires at least three APs with known positions to estimate the user's position.
- 3) Particle filter (PF) [17], [39] is a traditional recursive estimation method for indoor positioning based on Markov chains theory. It utilizes the posterior estimate from the previous time step to enhance the current positioning result.
- 4) Gaussian process regression (GPR) [17], [40] is a non-parametric regression method to generate a predicted fingerprint database with minimal effort. This workload-efficient fingerprinting method is compared to evaluate the positioning performance of the proposed method under unsupervised conditions.
- 5) Scaling by majorizing complex functions (SMACOF) [41] is a multidimensional scaling algorithm. It iteratively optimizes complex objective functions to achieve the optimal solution in a high-dimensional space. It is used as an iterative optimization positioning method to compare with our proposed geometry-based positioning method.



|                   | <span style="color: red;">F</span> fail | <span style="color: yellow;">✓</span> not robust | <span style="color: green;">✓</span> accurate |                                        |                                              |                                                |          |            |
|-------------------|-----------------------------------------|--------------------------------------------------|-----------------------------------------------|----------------------------------------|----------------------------------------------|------------------------------------------------|----------|------------|
|                   | <span style="color: red;">F</span> NC   | <span style="color: yellow;">✓</span> TRI        | <span style="color: yellow;">✓</span> PF      | <span style="color: red;">F</span> GPR | <span style="color: yellow;">✓</span> SMACOF | <span style="color: green;">✓</span> VM (ours) | end time |            |
| Narrow Corridor   |                                         |                                                  |                                               |                                        |                                              |                                                |          |            |
| Spacious Corridor |                                         |                                                  |                                               |                                        |                                              |                                                |          |            |
| Room              |                                         |                                                  |                                               |                                        |                                              |                                                |          |            |
|                   |                                         |                                                  |                                               |                                        |                                              |                                                |          | start time |

Fig. 10. Positioning results of various algorithms in different environments. The blue points denote the ground truth trajectories, and the red points indicate the estimated positions. The color gradient from light to dark indicates the time progression from start to finish. The cyan lines represent walls. The green triangles represent APs. The colored squares indicate the relative evaluations of the algorithms: red squares signify that positioning is difficult to implement, yellow squares indicate that positioning is possible but lacks robustness, and green squares represent that positioning can be achieved with relative accuracy.

6) Visibility Matching (VM, proposed method) uses visibility constraints of the AP to mitigate the impact of outliers on the estimated locations.

Outliers of NC are more pronounced compared to other methods because there is NC for RTT. When the number of APs is insufficient, TRI lacks anti-interference ability. In this study, the number of APs equals 3, and the positioning results are greatly affected by the outliers of any single AP ranging. PF can obtain smooth positioning results, but the accuracy is not much improved compared to TRI. This is because the likelihood update in the PF relies on the identical RTT measurement in TRI. The positioning results obtained through GPR fingerprinting are not as accurate when compared to other methods. This is due to the changes in AP visibility that lead to the non-Gaussian distribution between the true coordinates and the RTT measurements in the training data. Furthermore, for GPR, when the test data contains positions that are far from the training data coverage, the performance will significantly decline, primarily due to the model's insufficient modeling of these positions and the increased prediction uncertainty. SMACOF is an algorithm for multidimensional scaling, it achieves higher positioning accuracy than the previously mentioned approach but is also sensitive to outliers. VM achieves the highest positioning accuracy compared to other methods. In the spacious corridor experiment, the positioning results of all algorithms except for VM show obvious outliers, with the largest exceeding 6 m. The main reason is that AP2 and AP3 are placed close to each other, the device may only be able to rely on a few APs for positioning, which reduces the number of equivalent APs and affects the accuracy of positioning. By contrast, the positioning result of VM closely matches the ground truth trajectory and minimizes the presence of outliers.

The candidate score map in Fig. 11 illustrates the mechanism behind visibility matching. Different colors in the score map represent the possibility of a location being the

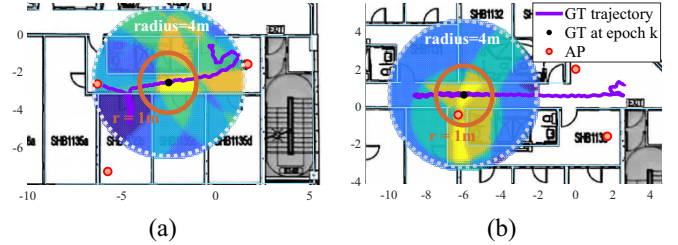


Fig. 11. Candidates score map within a 4 m radius in different experiments. Light yellow areas represent high scores, while deep blue areas represent low scores. The purple line indicates the ground truth trajectory, the black point denotes the true position of the user at epoch  $k$ , and the red points represent APs. The green triangles represent the locations of APs. (a) Candidate score map of the proposed method in suite experiment ( $k = 300$ ). (b) Candidate score map of the proposed method in narrow corridor experiment ( $k = 400$ ).

user's true position. In Fig. 11(a), we can observe candidates' scores within a 4 m radius of the user. Visibility matching distinguishes between enclosed rooms and corridors within a suite, restricting the highest scoring area to the corridor. In Fig. 11(b), multiple highest possibility areas impair positioning accuracy when the user is close to AP3. The reason is related to the relative position of the user to AP and the selection of RTT constraint tolerance (see in (9)). On the one hand, when the user passes near AP3, the RTT constraints imposed by other APs overlap with the RTT constraints of AP3 to a large extent; on the other hand, the choice of tolerance  $g$  in (9) will also lead to multiple high possibility areas. When the centers and radius of multiple rings are fixed, an increase in the width, i.e., tolerance, of the rings tends to form a continuous overlap region rather than multiple separate overlapping parts. Certainly, as tolerance increases, we must consider the influence of the introduced error. In practical applications, the radius of the sampling area must be carefully chosen; a larger radius can introduce more errors, and a smaller radius cannot match the user's motion state.

TABLE I  
COMPARISON OF POSITIONING ACCURACY AMONG DIFFERENT METHODS

| methods   | corridor      |               |               | lounge        |               |               | suite         |               |               |
|-----------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|           | mean          | median        | std           | mean          | median        | std           | mean          | median        | std           |
| NC        | 3.1831        | 3.4325        | 1.2867        | 4.5090        | 4.8745        | 1.3393        | 1.5553        | 1.5979        | 1.0278        |
| TRI       | 1.3990        | 1.1873        | 0.7330        | 2.4145        | 2.6827        | 1.0654        | 1.6724        | 1.6879        | 0.1720        |
| PF        | 1.4618        | 1.3139        | 0.7645        | 2.4444        | 2.5881        | 1.1110        | 1.8543        | 1.8661        | 0.3745        |
| GPR       | 6.6500        | 8.0523        | 2.3126        | 4.9650        | 5.9402        | 2.4983        | 8.8215        | 9.9450        | 2.2886        |
| SMACOF    | 1.4015        | 1.1661        | 0.8155        | 1.6009        | 1.4732        | 0.9461        | 1.8144        | 1.7722        | 0.1664        |
| <b>VM</b> | <b>0.6810</b> | <b>0.6728</b> | <b>0.2826</b> | <b>0.8012</b> | <b>0.7849</b> | <b>0.3529</b> | <b>0.6951</b> | <b>0.8232</b> | <b>0.3705</b> |

Table I presents the comparison results of all the methods. Among all the outcomes, the VM method demonstrates the highest accuracy in terms of mean, median, and standard deviation values. The advantages of VM can also be demonstrated by the CDF and positioning error in experiments, as shown in Fig. 12. The performance of NC and GPR is inferior to the other methods, with positioning errors greater than 5 m. TRI and PF cannot avoid the degradation of positioning accuracy caused by range measurement errors in NLOS environments. SMACOF achieves higher positioning accuracy than the previously mentioned methods. However, in the dynamic spacious corridor experiment where APs were placed adjacent to each other, the error also exceeds 5 m. For VM, nearly all positioning errors are below 4 m. Short-term ranging errors significantly impact other positioning methods but have minimal effect on VM. Comparison with PF reveals that the motion constraint of  $v \leq 1$  m/s is not the direct reason for improved positioning accuracy. PF employs similar motion constraints, and the positioning accuracy is not significantly improved compared with TRI. The most notable advantage of VM lies in the additional constraints provided by visibility matching, which avoids the damage of outlier NLOS measurements to positioning accuracy. In the proposed method in this article, positioning constraints are applied from two aspects: 1) ranging and 2) visibility matching. This approach fully utilizes the positioning features and mitigates the ranging errors introduced by NLOS reception, thereby enhancing positioning accuracy and robustness in indoor environments with NLOS receptions.

## VI. CONCLUSION

This article presents an RTT-based indoor positioning method with NLOS receptions. This method mitigates the ranging impact caused by NLOS reception and utilizes the NLOS feature of the signal to assist in positioning. First, the RTT multiwall ranging model is proposed to mitigate the RTT ranging errors. This model accounts for error sources when the signal passes through a wall, including the dielectric altering and the additional propagation distance introduced by refraction. Then, a method of visibility matching constraints is proposed. This method determines the AP visibility based on the strength of the received signal. It then uses areas with consistent AP visibility in the simulated visibility map as constraints for the solution. The advantage of the visibility map is that it creates a fingerprint database in a fixed room

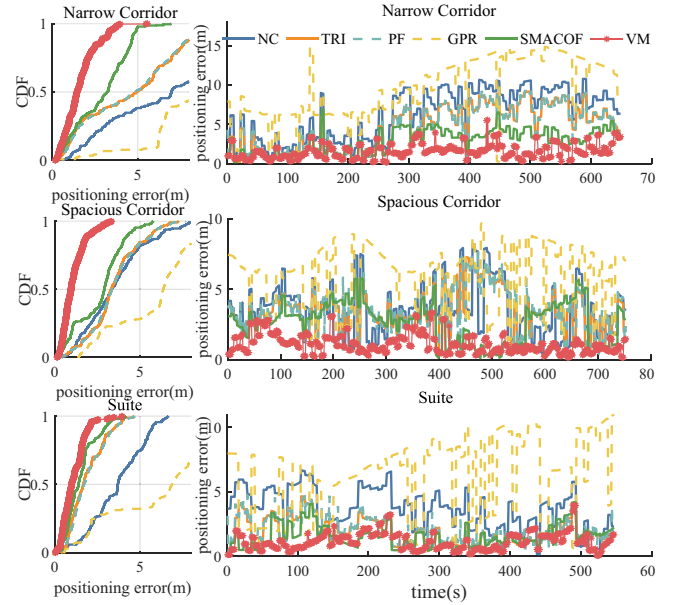


Fig. 12. Comparison of CDF and positioning errors across different algorithms in three experiments. The light blue, orange, cyan-blue, yellow, green, and red lines represent the positioning error results for NC, TRI, PF, GPR, SMACOF, and VM, respectively.

based on visibility features that do not need to be updated. Next, we propose a strategy that uses refined RTT ranging and visibility matching constraints to identify the user position that maximizes a specific objective function score. Experiment results in different environments demonstrate the accuracy and robustness of our method. The proposed method improves median positioning accuracy from 2.3 to 0.64 m compared to other methods. It effectively addresses the positioning challenges presented by harsh indoor NLOS conditions and a limited number of APs, thereby improving the overall accuracy of the positioning system.

The visibility matching can be extended for all radio-based technology in indoor positioning. However, this method may not achieve further precision in positioning under identical visibility conditions. For example, if a user is in a closed room, all AP signals are NLOS receptions for the user. Visibility matching can not determine which closed room the user is in. Our future research will focus on solving this limitation from the following aspects: 1) we will model the signal propagation path using ray tracing, estimating the user's position based on the possibility of each candidate and 2) we will improve

the accuracy by developing a more accurate error model and incorporating other feature measurements such as AOA and CSI.

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